

DeepSeek GPT: Understanding the Backend Workflow of a Modern AI Reasoning Model

1. Introduction

What is DeepSeek?

DeepSeek is an AI development firm based in Hangzhou, China. The company was founded by Liang Wenfeng, a graduate of Zhejiang University, in May 2023. Wenfeng also co-founded High-Flyer, a China-based quantitative hedge fund that owns DeepSeek. Currently, DeepSeek operates as an independent AI research lab under the umbrella of High-Flyer. The full amount of funding and the valuation of DeepSeek have not been publicly disclosed.

DeepSeek focuses on developing open source LLMs. The company's first model was released in November 2023. The company has iterated multiple times on its core LLM and has built out several different variations. However, it wasn't until January 2025 after the release of its R1 reasoning model that the company became globally famous.

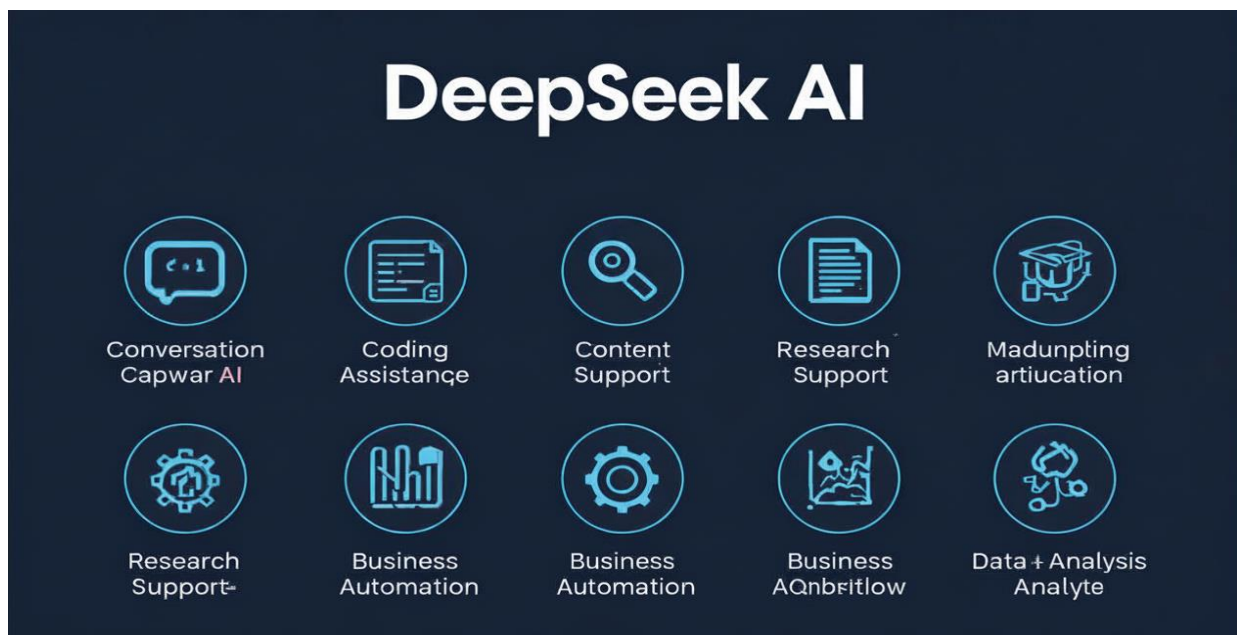
Purpose of DeepSeek

The main purpose of DeepSeek is to help people access information, solve problems, and complete tasks more efficiently. It is designed to understand user questions, provide meaningful answers, assist with learning, generate content, support coding, and simplify complex work. By using advanced AI technology, DeepSeek acts as a smart assistant that helps users save time, improve productivity, and make better use of information in their daily and professional activities.

Why Was It Created?

DeepSeek was created to make advanced artificial intelligence more efficient, accessible, and useful for everyone. The goal was to develop an AI system that could understand human language, solve complex problems, assist with coding, and provide intelligent responses while using resources more effectively. Its creators wanted to build a powerful AI model that could compete with leading AI technologies and help people learn, work, and make decisions more efficiently in their everyday lives. Where Is It Used?

DeepSeek is used in:



2. Problem It Solves

In today's fast-paced world, people often spend a lot of time searching for information, creating content, solving technical problems, and analyzing data. DeepSeek helps simplify these tasks by providing quick and intelligent assistance.

Why is this model important?

- Automation: Reduces repetitive manual work and improves efficiency.
- ☐ Content Generation: Helps create articles, reports, summaries, and other written content.
- Coding Assistance: Supports developers by generating code, finding bugs, and explaining programming concepts.
- Research Assistance: Quickly analyses large amounts of information and provides concise summaries.

3. Backend Workflow of DeepSeek

This section explains how DeepSeek processes a user query internally.

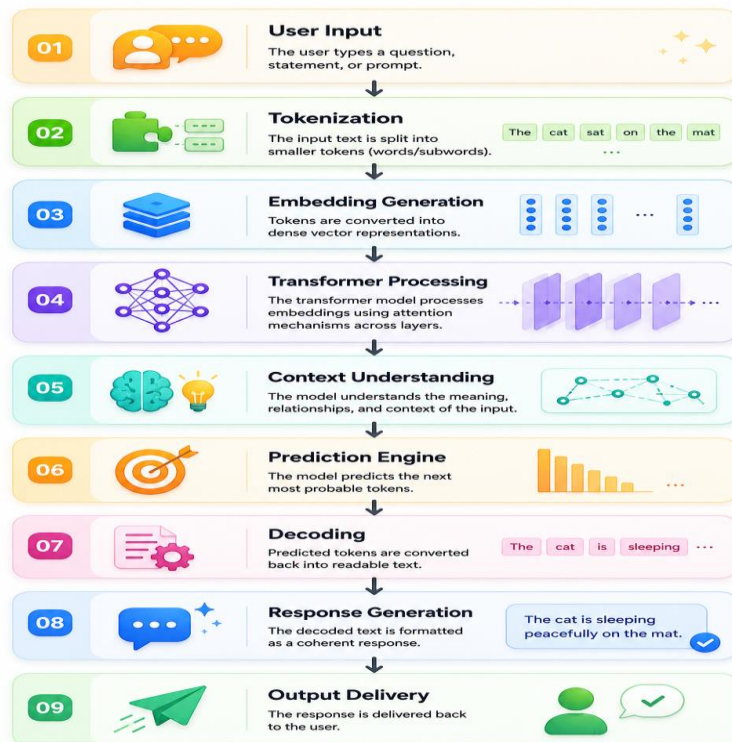
Step 1: User Input

The process starts when a user enters a question, command, or prompt. This input is sent to the AI system for processing.

Example:

"How does machine learning work?"

Flowchart 1: Overall Workflow



Step 2: Tokenization

Computers cannot understand language directly. Therefore, the input sentence is divided into smaller pieces called tokens. These tokens may be words, parts of words, or symbols.

Example:

Sentence: *"Artificial Intelligence is amazing."*

Tokens: *Artificial | Intelligence | is | amazing*

Each token is then assigned a numerical value that the model can process.

Step 3: Embedding Generation

After tokenization, the tokens are transformed into numerical vectors known as embeddings. These vectors capture the meaning and relationships between words.

Think of embeddings as a way of placing words in a mathematical space where words with similar meanings are located close to each other.

Benefits of embeddings:

- Better understanding of context
- Improved search and recommendation systems
- More accurate language processing

Step 4: Transformer Processing

This stage acts as the brain of the model. The transformer analyzes all tokens together and studies how they are related.

Attention Mechanism

The attention mechanism helps the model focus on the most important words in a sentence.

Example:

"The cat sat on the mat because it was tired."

The model identifies that the word *"it"* refers to *"the cat"*.

Understanding Context

Instead of reading words one by one, the transformer examines the entire sentence simultaneously, helping it understand meaning more effectively.

Pattern Learning

Through extensive training, the model learns patterns related to:

- Grammar
- Programming languages
- Mathematics
- Logical reasoning

Step 5: Context Understanding

The model keeps track of previous information within a conversation. This memory helps maintain continuity and provide relevant responses.

Example:

User: *"My favourite programming language is Python."*

Later: *"Suggest some useful libraries."*

The model understands that the user is asking about Python libraries.

Step 6: Prediction Engine

Rather than retrieving answers from a database, the AI predicts the most likely next word based on the given context.

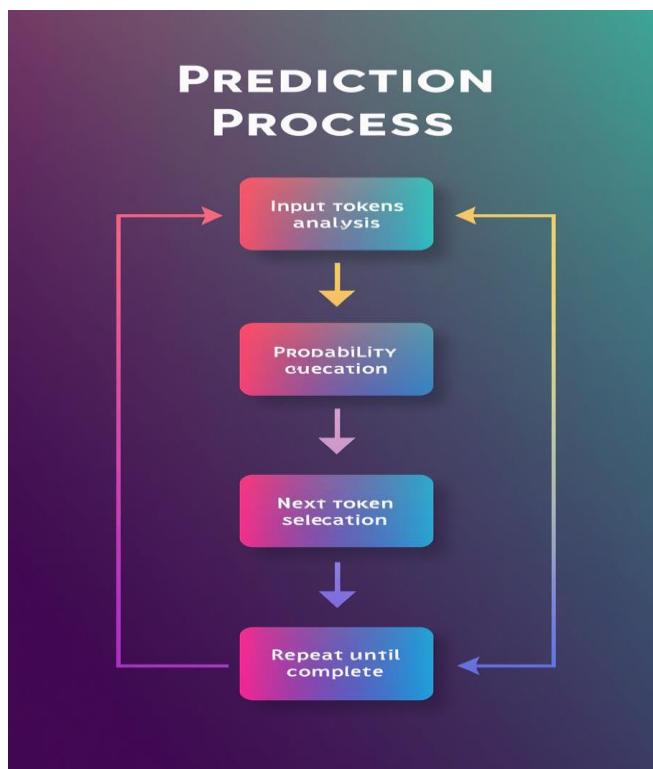
Example:

Input: *"The capital of France is"*

Predicted word: *"Paris"*

The model calculates probabilities for many possible words and selects the most suitable option.

Pridiction Process of DeepSeek



Step 7: Decoding

The predicted tokens are converted into readable text. Different decoding methods help balance creativity and accuracy.

- Low temperature: Produces more factual and predictable responses.
- High temperature: Generates more creative and varied outputs.

This stage ensures that the response sounds natural and coherent.

Step 8: Response Generation

The selected words are combined into meaningful sentences and paragraphs.

Example:

Generated response:

"Machine learning is a branch of artificial intelligence that enables computers to learn from data and improve their performance over time."

Step 9: Output Delivery

Finally, the completed response is sent back to the user through the interface. The entire process happens within seconds due to powerful computing resources and optimized algorithms.

4. Unique Features of DeepSeek

Several features distinguish DeepSeek from other AI models.

Better Reasoning

DeepSeek-R1 is specifically trained to perform logical reasoning and complex problem solving.

Mixture of Experts (MoE)

Instead of activating all parameters, DeepSeek activates only relevant expert networks, improving efficiency.

Multi-Head Latent Attention (MLA)

Reduces memory usage while preserving contextual understanding.

Cost Efficiency

DeepSeek achieves competitive performance while reducing training and inference costs.

Coding and Mathematics Strength

DeepSeek performs exceptionally well in programming and mathematical reasoning tasks.

5. Real-World Applications

DeepSeek is being used in many areas to make work faster, easier, and more efficient. Its ability to understand language, generate content, and solve problems makes it useful for both individuals and organizations.

1. Customer Support and Chatbots

Many companies use AI-powered chatbots to answer customer questions instantly. DeepSeek can provide quick responses, solve common issues, and assist users at any time of the day, reducing the workload on human support teams.

2. Software Development

Programmers can use DeepSeek as a coding assistant. It helps by suggesting code, finding errors, explaining programming concepts, and improving code quality. This allows developers to complete projects more efficiently.

3. Education and Learning

Students and teachers can benefit from DeepSeek in various ways. It can explain difficult topics, create study notes, answer questions, and provide personalized learning support. This makes learning more interactive and accessible.

4. Content Creation

Writers, marketers, and content creators can use DeepSeek to generate ideas, draft articles, create reports, and prepare social media content. It helps save time while maintaining productivity.

5. Healthcare Support

In the healthcare sector, DeepSeek can assist in summarizing medical documents, organizing patient information, and helping professionals access relevant information more quickly. However, human experts are still required for final medical decisions.

6. Financial Services

Banks and financial institutions can use DeepSeek to analyze data, generate reports, answer customer inquiries, and assist in decision-making processes. This helps improve efficiency and accuracy.

7. Research and Data Analysis

Researchers often need to review large amounts of information. DeepSeek can summarize lengthy documents, identify key points, and help organize findings, allowing researchers to focus on deeper analysis.

8. Business Operations

Organizations use DeepSeek to automate routine tasks such as report writing, email drafting, document management, and information retrieval. This increases productivity and reduces manual effort.

6. Limitations

◆ Hallucinations

Sometimes DeepSeek may provide information that sounds completely correct but is actually inaccurate. This happens because the model generates responses based on patterns rather than verifying facts.

◆ Bias in Outputs

The model learns from large amounts of internet and text data. If that data contains biases, some of those biases may occasionally appear in its responses.

◆ Limited Context Understanding

DeepSeek can remember a lot of information within a conversation, but extremely long discussions or documents may cause it to lose track of certain details.

◆ High Resource Requirements

Advanced AI models require powerful computers and significant processing power. This makes them expensive to run and maintain.

◆ **Prediction-Based Responses**

DeepSeek does not truly understand information the way humans do. It predicts the most likely answer based on patterns it has learned, which can sometimes result in mistakes.

◆ **Reasoning Challenges**

Although DeepSeek is designed to handle logical reasoning, it can still make errors when faced with highly complex problems or situations that require deep human judgment

7. Conclusion

DeepSeek represents a significant step forward in the field of artificial intelligence. Its ability to understand language, generate content, assist with coding, and solve complex problems makes it a valuable tool for individuals and organizations alike. By combining advanced technologies such as transformers, reasoning models, and efficient processing techniques, DeepSeek can deliver fast and context-aware responses across a wide range of applications.

However, like all AI systems, it is not perfect. It can occasionally make mistakes, generate inaccurate information, or struggle with highly complex situations. Therefore, its outputs should be used as support rather than as a complete replacement for human expertise and judgment.

As AI technology continues to evolve, DeepSeek is expected to become even more capable and efficient. With responsible use and continuous improvement, it has the potential to play an important role in shaping the future of education, research, business, and many other fields.